



phere and onto the runway but also in the electronic components integrated in the space vehicle design. Electronic component suppliers for space electronics must meet stringent temperature standards, along with radiation-hardening specs. These requirements will only get tougher to meet further requirements as scientists plan to send humans beyond the low-earth orbit to explore the solar system.

Faster turnaround times for satellites and other launches are narrowing the delivery window for aerospace electronics, putting added pressure on electronics technology. With greater requirements for testing space electronics ranging from thermal performance to radiation dose and production tests, the pendulum is swinging away from using commercial parts in space. Further, the requirement for space electronics for use in deep-space probes or in instrumentation to be used on scientific payloads such as earth-imaging and weather satellites is increasing.

Recently, a spacecraft bound for Jupiter got an armour suit to protect it from the fiercest radiation any space probe has ever encountered. The unmanned Juno space probe will

face a treacherous environment with more radiation than around any other planet and needs an armoured shield around its sensitive electronics. Juno is basically an armoured tank going to Jupiter; without its protective shield or radiation vault, its electronics would damage on the very first pass near Jupiter.

The six-sided radiation shield is made of titanium. Lead, an effective shield against radiation, would be too soft to survive vibrational forces and stresses. While the vault is not designed to completely prevent Jupiter's radiation from hitting the system, it is expected to dramatically slow down the ageing effect that radiation has on electronics for the duration of the mission.

Problems with space electronics

Reliability of components. Space agencies place reliability at the top of their priorities since the failure of just one component can lead to the loss of a multi-million dollar mission. A clear counter trend is the use of commercial off-the-shelf (COTS) components. While these parts are generally more advanced in terms of processor perfor-

mance and logic or memory density than those designed specifically for use in military or space borne systems, COTS devices do not have the background of design and extensive testing that ensures reliability.

Parts designed for military and aerospace often have a wider range of materials at their disposal than COTS components, particularly when it comes to the materials used to connect the chips to the printed control boards. The more restricted materials used in COTS tend to suffer from greater reliability problems under heat stress. Military-grade parts used for space flight are also tested to a much greater degree than their COTS counterparts—not just over a wider temperature range but also with a greater concentration on faults that cannot be picked up by generic wafer probe and package level tests.

Radiation effect. When a space-bound electronic component passes its test, there remains one big problem—radiation. Radiation is one of the main characteristics of space weather. Radiations of galactic and solar origin determine radiation hazards for people and technology, computer and memory upsets and failures, solar cell damage, radio wave propagation disturbances, and failures in communication and navigation systems.

The effect of ionising radiation on hard-wired logic circuits is less pronounced. These errors are typically transient and often non-destructive. A review conducted by NASA in 1996 of a hundred failures and problems on its spacecraft found that one-third of the failures were caused by ionising radiation leading to single event upsets (state changes in logic or memory) or permanent degradation in the performance of onboard electronic devices. Sometimes these single event upsets are even capable of destroying computer memories on the earth. But obviously with a much larger probability in spacecraft systems during periods of large energetic particle fluxes, it is advisable to switch off some part of the electronics to protect